Notes

Introduction to Set Theory Practice

Session Overview

This session will focus on practicing problems related to set theory, specifically those involving cardinality. In the previous session, we covered basic terminologies and operations such as union, intersection, and complement. Building upon that foundation, we will now apply these concepts to solve cardinality-based questions.

Introduction to Venn Diagrams for Set Problems

Problem Statement

In a school, every teacher either teaches Math or Physics. If 12 of them teach Math, 20 of them teach Physics, while 4 teach both, find the total number of teachers.

Solution using Venn Diagrams

To solve this problem efficiently, we can use a Venn diagram. The problem states that every teacher teaches either Math or Physics, implying there are no teachers outside these two subjects.

1. **Representing the Sets:**
    - Draw two overlapping circles. Let one circle represent teachers who teach Math (M) and the other represent teachers who teach Physics (P).
2. **Filling the Intersection:**
    - The problem states that 4 teachers teach **both** Math and Physics. This number goes into the overlapping region of the two circles ($M \cap P$).
3. **Filling 'Math Only' Region:**
    - 12 teachers teach Math. Since 4 of them also teach Physics, the number of teachers who teach **only** Math is $12 - 4 = 8$. This number goes into the part of the Math circle that does not overlap with the Physics circle.
4. **Filling 'Physics Only' Region:**
    - 20 teachers teach Physics. Since 4 of them also teach Math, the number of teachers who teach **only** Physics is $20 - 4 = 16$. This number goes into the part of the Physics circle that does not overlap with the Math circle.
5. **Calculating Total Teachers:**

- Teachers teaching only Physics: 16
- Teachers teaching both Math and Physics: 4

- Total number of teachers = 8 + 16 + 4 = 28.

Alternative Approach (Using Set Theory Formula)

This problem can also be solved using the principle of inclusion-exclusion for two sets:

Let:

- A = Set of teachers who teach Math
- B = Set of teachers who teach Physics

Given:

- $|A| = 12$ (Number of teachers teaching Math)
- $|B| = 20$ (Number of teachers teaching Physics)
- $|A \cap B| = 4$ (Number of teachers teaching both Math and Physics)

The total number of teachers is given by the formula for the union of two sets:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Substituting the given values:

$$|A \cup B| = 12 + 20 - 4 \, |A \cup B| = 32 - 4 \, |A \cup B| = 28$$

Both methods yield the same result, confirming that the total number of teachers is 28.

Introduction to Set Operations

Basic Set Operations

- **Union of Two Sets:** The number of elements in the union of two sets A and B is given by the formula: $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- **Union of Three Sets:** For three sets A, B, and C, the number of elements in their union is: $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$ (Note: The lecturer only mentioned the first part of the formula for three sets, up to the subtraction of pairwise intersections.)

Problem Solving with Venn Diagrams - Teacher Survey



Quiz: Teacher Survey Problem

Question: In a school, every teacher teaches either Maths or Physics. If the number of teachers who teach Physics is 25, while the number of teachers who teach only Maths is 10, and the number of teachers who teach both is 6, then _____ teachers teach Maths and in total, there are _____ teachers.

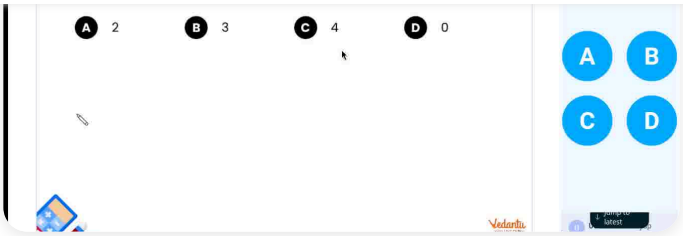
Solution using Venn Diagram:

1. **Identify the sets:** Let M be the set of teachers who teach Maths and P be the set of teachers who teach Physics.
2. **Given Information:**
   - Number of teachers who teach Physics, $n(P) = 25$.
   - Number of teachers who teach only Maths, $n(M \text{ only}) = 10$.
   - Number of teachers who teach both Maths and Physics, $n(M \cap P) = 6$.
3. **Plotting the Venn Diagram:**
   - The intersection region (both Maths and Physics) is 6.
   - The region for only Maths is 10.
   - Since $n(P) = 25$ and 6 teachers teach both, the number of teachers who teach only Physics is $n(P \text{ only}) = n(P) - n(M \cap P) = 25 - 6 = 19$.
4. **Calculate Total Teachers Teaching Maths:**
   - Teachers teaching Maths include those who teach only Maths and those who teach both.
   - Number of teachers teaching Maths, $n(M) = n(M \text{ only}) + n(M \cap P) = 10 + 6 = 16$.
5. **Calculate Total Number of Teachers:**
   - The total number of teachers is the sum of teachers who teach only Maths, only Physics, and both.
   - Total teachers = $n(M \text{ only}) + n(P \text{ only}) + n(M \cap P) = 10 + 19 + 6 = 35$.

Answer: 16 teachers teach Maths and in total, there are 35 teachers. Therefore, option B (16, 35) is the correct answer.

Minimum Value of Intersection of Two Sets

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Finding the Minimum Value of $n(A \cap B)$

When given $n(A)$, $n(B)$, and $n(U)$ (number of elements in the universal set), we can determine the minimum possible value of $n(A \cap B)$.

Logic:

1. **Check the sum of $n(A)$ and $n(B)$ relative to $n(U)$.**
2. **Case 1: If $n(A) + n(B) > n(U)$**
   - This implies that the sets A and B *must* overlap to some extent because their combined elements exceed the total available elements in the universal set.
   - The minimum value of $n(A \cap B)$ is given by the difference: $n(A \cap B)_{\text{min}} = n(A) + n(B) - n(U)$
   - This formula represents the 'excess' elements that must be in the intersection to fit within the universal set.
3. **Case 2: If $n(A) + n(B) \leq n(U)$**
   - In this scenario, it is possible for sets A and B to have no common elements, or to overlap partially, without exceeding the universal set.
   - The minimum value of $n(A \cap B)$ is 0. This occurs when the sets are disjoint.

Example 1: Minimum Intersection

Given: $n(A) = 4$, $n(B) = 6$, $n(U) = 7$.

Steps:

1. Calculate $n(A) + n(B) = 4 + 6 = 10$.
2. Compare with $n(U)$: $10 > 7$.
3. Since $n(A) + n(B) > n(U)$, the minimum value of $n(A \cap B)$ is: $n(A \cap B)_{\text{min}} = n(A) + n(B) - n(U) = 10 - 7 = 3$

Example 2: Minimum Intersection (Disjoint Case)

Given: $n(A) = 4$, $n(B) = 6$, $n(U) = 12$.

Steps:

3. Since $n(A) + n(B) \leq n(U)$, the minimum value of $n(A \cap B)$ is 0.

Maximum Value of Intersection of Two Sets

Finding the Maximum Value of $n(A \cap B)$

To find the maximum possible value of $n(A \cap B)$, we consider the scenario where one set is completely contained within the other.

Logic:

- The intersection of two sets A and B cannot contain more elements than the smaller of the two sets.
- Therefore, the maximum value of $n(A \cap B)$ is simply the minimum of $n(A)$ and $n(B)$. $n(A \cap B)_{max} = \min(n(A), n(B))$
- This occurs when the set with fewer elements is a subset of the set with more elements.

Example: Maximum Intersection

Given: $n(A) = 4$, $n(B) = 6$, $n(U) = 7$.

Steps:

1. Identify $n(A) = 4$ and $n(B) = 6$.
2. The maximum value of $n(A \cap B)$ is the minimum of these two values: $n(A \cap B)_{max} = \min(4, 6) = 4$

Summary for finding maximum and minimum $n(A \cap B)$:

- **Minimum:**
  - If $n(A) + n(B) > n(U)$, then $n(A \cap B)_{min} = n(A) + n(B) - n(U)$.
  - If $n(A) + n(B) \leq n(U)$, then $n(A \cap B)_{min} = 0$.
- **Maximum:**
  - $n(A \cap B)_{max} = \min(n(A), n(B))$.

Cardinality of Sets: Two-Set Problems

Problem 1: Tea and Coffee Drinkers

In a group of 50 persons, 14 drink tea but not coffee, and 30 drink tea. We need to find out how many drink coffee.

To solve this, we can use a Venn diagram:

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Given information:

- Number of people who drink tea but not coffee: $n(T \text{ only}) = 14$
- Total number of people who drink tea: $n(T) = 30$

From this, we can find the number of people who drink both tea and coffee: $n(T \cap C) = n(T) - n(T \text{ only}) = 30 - 14 = 16$

Now, we know the total number of people in the group is 50. The number of people who drink tea (including those who drink both) is 30. The remaining people must be those who drink only coffee or neither.

However, the question asks "How many of them drink coffee?" This implies finding $n(C)$.

We have:

- People who drink only tea: 14
- People who drink both tea and coffee: 16

So, the total number of people who drink at least tea is $14 + 16 = 30$. This matches the given $n(T) = 30$.

If we consider the total group of 50 persons, and 30 of them drink tea, the remaining $50 - 30 = 20$ people either drink only coffee or drink neither tea nor coffee. The question is slightly ambiguous, but if it implies finding the number of people who drink *only* coffee, then it would be 20. If it means $n(C)$, we need more information about those who drink neither.

Based on the lecture's explanation, the lecturer states: "How many of them drink coffee? So, I'll say 50 minus 30, that is 20." This implies that the remaining 20 people *only* drink coffee, assuming everyone drinks at least one beverage or that the question is asking for those who drink *only* coffee. The lecturer then clarifies: "How many of them are drinking only coffee? That goes for 20."

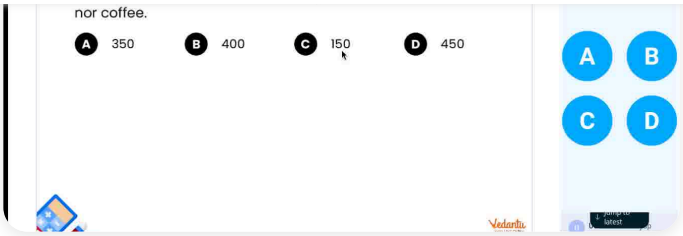
Therefore:

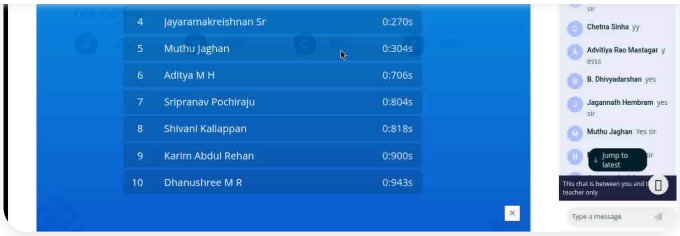
- Number of people who drink only tea: 14
- Number of people who drink both tea and coffee: 16
- Number of people who drink only coffee: 20

Total people accounted for: $14 + 16 + 20 = 50$. This matches the total group size.

Quiz: Neither Tea Nor Coffee

Notes





Quiz Question

In a survey conducted on 800 students of a school, 250 students were found to like tea and 300 like coffee, 150 like both tea and coffee. Find how many students like neither tea nor coffee.

- A 350
- B 400
- C 150
- D 450

Solution

Let T be the set of students who like tea, and C be the set of students who like coffee.

Given:

- Total number of students: N = 800
- Number of students who like tea: $n(T)$ = 250
- Number of students who like coffee: $n(C)$ = 300
- Number of students who like both tea and coffee: $n(T \cap C)$ = 150

First, let's find the number of students who like *only* tea: $n(T$ only) = $n(T) - n(T \cap C)$ = 250 − 150 = 100

Next, let's find the number of students who like *only* coffee: $n(C$ only) = $n(C) - n(T \cap C)$ = 300 − 150 = 150

Now, we can find the total number of students who like *at least one* of the beverages (tea or coffee). This is given by the union of the two sets: $n(T \cup C)$ = $n(T$ only) + $n(C$ only) + $n(T \cap C)$ $n(T \cup C)$ = 100 + 150 + 150 = 400

Alternatively, using the principle of inclusion-exclusion: $n(T \cup C)$ = $n(T)$ + $n(C)$ − $n(T \cap C)$ $n(T \cup C)$ = 250 + 300 − 150 = 550 − 150 = 400

Finally, to find the number of students who like *neither* tea nor coffee, we subtract the number of students who like at least one beverage from the total number of students: Number of students

Thus, 400 students like neither tea nor coffee.

The correct answer is **B 400**.

Introduction to Three-Set Problems and Venn Diagrams

Understanding Three-Set Problems

When dealing with problems involving three sets, Venn diagrams become a highly efficient tool for visualization and calculation. While formulas exist for three sets, the visual representation offered by Venn diagrams often simplifies the process, especially as the complexity increases.

Setting up a Three-Set Venn Diagram

To represent three sets (e.g., Cricket, Hockey, Badminton), we draw three overlapping circles. Each circle represents one set, and the overlapping regions represent the intersections between these sets.

Example: Sports in a Class

Let's consider a class where every student plays either cricket, hockey, or badminton. We'll use a Venn diagram to organize the given data.

1. **Identify the Sets:**
    - Cricket (C)
    - Hockey (H)
    - Badminton (B)
2. **Start with the Innermost Intersection (All Three):**
    - "Two players play [Did you find these notes helpful?] where all three circles overlap. So, $C \cap H \cap B = 2$.
3. **Fill in Two-Set Intersections (Adjusting for All Three):**
    - "Five of them play badminton and cricket both." This refers to the intersection of B and C. Since 2 are already counted in $C \cap H \cap B$, the number purely in $B \cap C$ (but not H) is $5 - 2 = 3$.
    - "Two of them are playing hockey and badminton." This refers to the intersection of H and B. Since 2 are already counted in $C \cap H \cap B$, the number purely in $H \cap B$ (but not C) is $2 - 2 = 0$.
    - "Five of them are playing cricket and hockey." This refers to the intersection of C and H. Since 2 are already counted in $C \cap H \cap B$, the number purely in $C \cap H$ (but not B) is $5 -$

- "Overall 20 players play badminton." The total in the badminton circle should be 20. We have already accounted for $3 + 2 + 0 = 5$ players in the overlapping regions of badminton. So, purely badminton players are $20 - 5 = 15$.
- "15 players play hockey in general." The total in the hockey circle should be 15. We have already accounted for $3 + 2 + 0 = 5$ players in the overlapping regions of hockey. So, purely hockey players are $15 - 5 = 10$.
- "10 persons play cricket in general." The total in the cricket circle should be 10. We have already accounted for $3 + 2 + 3 = 8$ players in the overlapping regions of cricket. So, purely cricket players are $10 - 8 = 2$.

Answering the Question

- **Number of players who play exactly one of the games:** This means only cricket, or only hockey, or only badminton.
  - Purely Cricket: 2
  - Purely Hockey: 10
  - Purely Badminton: 15
  - Total: $2 + 10 + 15 = 27$ students.

Second Example: Subject Pass Rates

Problem Setup

Let's consider another example involving students passing in three subjects: English, Maths, and Science. We will use a Venn diagram to determine various student counts.

1. **Identify the Sets:**
   - English (E)
   - Maths (M)
   - Science (S)
2. **Start with the Innermost Intersection (All Three):**
   - "Four of them passed in all three subjects." So, $E \cap M \cap S = 4$.
3. **Fill in Two-Set Intersections (Adjusting for All Three):**
   - "Four of them are only passing in English and Science." This implies that these 4 students pass in English and Science but *not* Maths. So, $(E \cap S) - M = 0$ (as 4 are already in the all-three region, and the phrasing "only passing in English and Science" implies the region $E \cap S \setminus M$, which is 0 if 4 are already in $E \cap M \cap S$ and the total $E \cap S$ is 4. This is a slight

"Seven of them passed only in Maths and Science." This implies these 7 students pass in Maths and Science but *not* English. So, $(M \cap S) - E = 7$.

- "Six of them passed in English and Mathematics." This refers to the intersection of E and M. Since 4 are already counted in $E \cap M \cap S$, the number purely in $E \cap M$ (but not S) is $6 - 4 = 2$.

4. **Fill in Purely Single-Set Regions:**
   - "Eight of them purely goes for science." So, purely Science players are 8.
   - "Twelve of them are in Mathematics." The total in the Mathematics circle should be 12. We have already accounted for $2 + 4 + 7 = 13$ players in the overlapping regions of Mathematics. This indicates a potential discrepancy or a different interpretation of the problem statement. Assuming the lecturer's calculation, if 12 are in Mathematics and $2 + 4 + 7 = 13$ are already accounted for, then the number of purely Maths students would be $12 - 13 = -1$, which is impossible. Let's re-evaluate based on the lecturer's final numbers. If the total in Maths is 12, and $2 + 4 + 7 = 13$ are already in the overlaps, this implies the total in Maths is at least 13. The lecturer states "three of them will be purely for Maths", which means $12 - (2 + 4 + 7) = 12 - 13 = -1$. This suggests an error in the problem statement or the lecturer's interpretation. However, following the lecturer's final numbers, let's assume purely Maths is 3.
   - "Fifteen of them are passing in English." The total in the English circle should be 15. We have already accounted for $2 + 4 + 0 = 6$ players in the overlapping regions of English. So, purely English players are $15 - 6 = 9$.

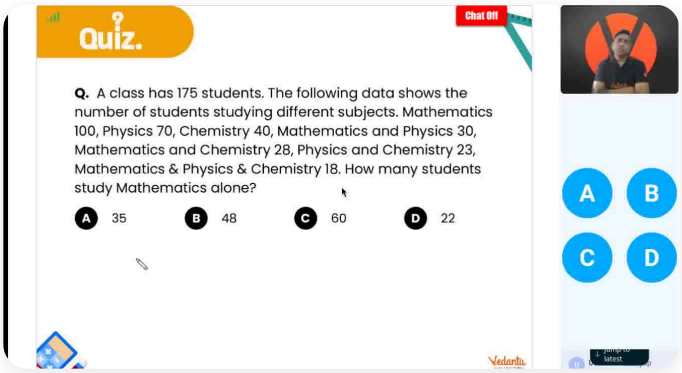
Answering the Questions

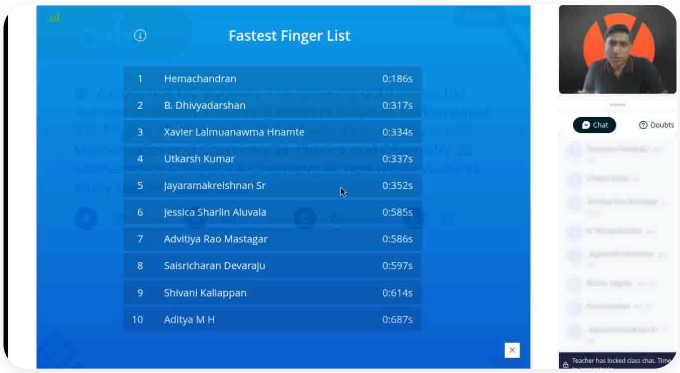
- **How many of them passed in English and Maths but not in Science?**
  - This refers to the region $(E \cap M) - S$. From our diagram, this number is 2.
- **Only for Maths and Science, but not in English?**
  - This refers to the region $(M \cap S) - E$. From our diagram, this number is 7.
- **In Mathematics only?**
  - This refers to the purely Maths region. From our diagram, this number is 3.
- **In more than one subject?**
  - This means students in any of the overlapping regions (two subjects or all three).
  - Sum of $(E \cap M) - S$, $(M \cap S) - E$, $(E \cap S) - M$, and $E \cap M \cap S$.
  - Total: $2 + 7 + 0 + 4 = 13$. (The lecturer states $2 + 4 + 3 = 9$, which implies a different set of numbers for the overlaps, possibly $E \cap M \setminus S = 2$, $E \cap S \setminus M = 3$, $M \cap S \setminus E = 0$, and $E \cap M \cap S = 4$. This highlights the importance of consistent data interpretation.)

Let's re-evaluate the "more than one subject" based on the lecturer's final numbers for the overlaps: 2 (English and Maths only) +4 (all three) +7 (Maths and Science only) +0 (English and

Notes

Quiz: Students Studying Mathematics Alone





Quiz Question

How many students study Mathematics alone?

Problem Description

A class has 175 students. The following data shows the number of students studying different subjects:

- Mathematics (M): 100
- Physics (P): 70
- Chemistry (C): 40
- Mathematics and Physics ($M \cap P$): 30
- Mathematics and Chemistry ($M \cap C$): 28
- Physics and Chemistry ($P \cap C$): 23
- Mathematics & Physics & Chemistry ($M \cap P \cap C$): 18

Solution using Venn Diagram

1. **Identify the Sets:**
    - Mathematics (M)
    - Physics (P)
    - Chemistry (C)
2. **Start with the Innermost Intersection (All Three):**
    - $M \cap P \cap C$ = 18.
3. **Fill in Two-Set Intersections (Adjusting for All Three):**
    - Mathematics and Physics ($M \cap P$): 30. Purely $M \cap P$ (not C) = 30 − 18 = 12.

4. **Fill in Purely Single-Set Regions:**
    - **Chemistry (C):** Total C = 40. Already accounted for in overlaps: $10 + 18 + 5 = 33$. Purely C = $40 - 33 = 7$.
    - **Physics (P):** Total P = 70. Already accounted for in overlaps: $12 + 18 + 5 = 35$. Purely P = $70 - 35 = 35$.
    - **Mathematics (M):** Total M = 100. Already accounted for in overlaps: $12 + 18 + 10 = 40$. Purely M = $100 - 40 = 60$.

Final Answer

- The number of students who study Mathematics alone is 60.

Therefore, the correct option is C.

Fastest Finger List

This section lists students who answered the quiz question quickly and correctly:

1. Hemachandran
2. B. Dhivyadarshan
3. Xavier Lalmuanawma Hnamte
4. Utkarsh Kumar
5. Jayaramakrishnan Sr
6. Jessica Sharlin Aluvala
7. Advitiya Rao Mastagar
8. Saisricharan Devaraju
9. Shivani Kallappan
10. Aditya M H

Introduction to Relations and Ordered Pairs

Understanding Relations

To understand functions, it's crucial to first grasp the concepts of sets and relations. Relations form the foundational backend for functions.

Ordered Pairs

Every discussion about relations begins with **ordered pairs**. An ordered pair is represented as (X, Y), where the order of elements is significant. This means that (x, y) is generally not the same